PAPER<i>340 ■ EDM SO(10) BSM Refined F</i>u Coupling: Darkonia Phase Boundary P_SCm = 1

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Session: *96*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST SO(10) EDM darkonia phase boundary in UQFF; FIRST V_cb coupling derivation **Author:** Daniel T. Murphy

<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

The UQFF F*u coupling is refined using SO(10) grand unification predictions for the electron electric dipole moment (EDM) d*e ~ 10?■5 e■cm. A new phase boundary is identified at P*SCm = 1 where darkonia (dark-sector charmonium analogs) become stable. The V*cb CKM element coupling k?■*GF*■■s/p and the tau-lepton anomaly deviation t_dev = 5■10?8 s (< 5% error relative to Super Tau-Charm factory limits) are derived.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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2. Core Physics

2.1 EDM-Enhanced F_u Coupling

The refined F_u positive coupling from SO(10) EDM:

```
F_u^+ = \frac{d_e \cdot e}{2 m_e c} \cdot e^{-[SSq] \cdot n / 26}
```

where:

- d_e = 1.6■10?44 C■m (SI equivalent of ~10?■5 e■cm SO(10) prediction)
- e = 1.6■10?■? C
- m_e = 9.11■10?■■ kg
- [SSq] = 0.57
- n = 13 (pseudo-scalar Ramanujan state; half-sum of 26 states)

At n = 13:

```
F_u^+(n=13) \approx 6.5 \times 10^{-45} \ \mathrm{N}
```

2.2 Darkonia Stable Phase Boundary

The UQFF superconductive modifier P_SCm = 1 defines the phase boundary at which dark-sector mesons ("darkonia") become kinematically stable against decay via the UQFF vacuum polarization channel:

```
P_{\rm SCm} = 1 \implies \text{darkonia stable}
```

This is analogous to the Meissner effect in the gravitational channel (PAPER_266), but applies to the BSM sector.

2.3 V_cb Coupling

The V_cb CKM matrix element coupling enters via:

```
V_{cb}^{\rm UQFF} = k_\eta \cdot G_F^2 \cdot s / \pi
```

where:

k_? = 10?■■ (UQFF aether coupling)

G_F = 1.1664■10?5 GeV?■ (Fermi constant)

s = centre-of-mass energy squared (GeV■)

V_cb = (40.5 ■ 1.3)■10?■ (PDG 2024)

2.4 tau_dev from g-2 UQFF Fit

From PAPER*333 BSM fit: a = 4.74■10?5, b = 9.96, ?*Higgs = 47.34 ? t_dev = 5■10?8 s, error < 5% compared to Super Tau-Charm factory precision target.

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3. Key Values

Quantity	Value	Notes
d_e (SO(10))	~10?■5 e■cm	3■ below ACME limit
F_u?(n=13)	6.5■10?45 N	[SSq] Ramanujan suppression
P_SCm phase boundary	P_SCm = 1	darkonia stable
V_cb (PDG)	40.5■10?■	CKM reference
t_dev	5■10?8 s	g-2 fit < 5% error
?_Higgs	47.34	Universal UQFF Higgs coupling

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4. Physical Significance

The detection of a non-zero d_e at the SO(10) level would simultaneously:

Confirm BSM CP violation at the TeV scale

Provide a V_cb■UQFF coupling constraint independent of lattice QCD

Validate the UQFF P_SCm = 1 darkonia phase boundary through missing-energy searches at BES-III or Super Tau-Charm

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5. Deduplication Note

PAPER_333 (Session 95): BSM 10-experiment package ■ bundled multi-experiment class

PAPER_340: Standalone SO(10) EDM refinement with darkonia phase boundary ■ **FIRST P_SCm=1 darkonia stability**

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6. Classification

Physics Territory: FIRST SO(10) EDM darkonia phase boundary + V_cb coupling in UQFF **Scale:** Particle physics (sub-fermi) **CP Implementation:** `EDMSO10BSMRefinedFuCalculator` (CondensedPhysics3.py, Session 96)
